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Machine Learning as a Service Cloud Selection: An MCDM Approach for Optimal Decision Making

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Abstract

Machine Learning as a Service, MLaaS refers to cloud-based platforms that provide machine learning tools and infrastructure to users, allowing them to access and utilize machine learning capabilities without managing the underlying hardware and software. Cloud service providers offer scalable and flexible resources, enabling businesses to train and deploy machine learning models efficiently and cost-effectively. Multi-Criteria Decision making, MCDM helps individuals or organizations make choices when multiple criteria or objectives are involved. It aims to find the best alternative among a set of available options by considering various criteria or factors that are relevant to the decision. Machine learning as a service (MLaaS) and MCDM are two distinct concepts that can be combined to create powerful decision-making solutions. The present paper proposed an integrated approach based on the Analytic Hierarchy Process and TOPSIS method for choosing the best MLaaS cloud. Four alternatives for MLaaS cloud were analyzed based on seventeen shortlisted criteria. Based on the calculations, conclusions have been reached and potential study areas have been identified.

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Keywords: Machine Learning as a Service (MLaaS); Multi Criteria Decision making (MCDM); Analytic Hierarchy Process (AHP); TOPSIS.

1. Introduction

Cloud computing refers to the delivery of computing services over the internet, enabling users to access and utilize a wide range of resources, such as virtual machines, storage, databases, and applications, without the need for

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on-site hardware or infrastructure. Machine Learning as a Service (MLaaS) is a cloud computing approach, that helps developers, data scientists, and businesses to access and use machine learning tools and resources over the internet without the need for extensive hardware and software infrastructure [1]. In the proposed study, various commercial Machine Learning as a Service over cloud service providers are analyzed and ranked from best to worst.

Amazon Web Services introduced Amazon SageMaker in 2017. SageMaker made it easier for developers and data scientists to access the necessary tools and resources for machine learning without the need for extensive infrastructure setup. Following AWS's lead, service providers like Microsoft Azure, Google Cloud Platform, and IBM Cloud have also introduced their own MLaaS offerings, contributing to the widespread adoption of machine learning as a service across the industry. There are many Cloud service providers providing Machine learning as a Service. However, major service providers in terms of their market share were selected in the proposed study. Each provider may have specific strengths and weaknesses depending on your particular use case and requirements. Thus the selection of the best cloud service provider, offering Machine Learning as a service can be addressed by the tools and techniques available under the domain of Multi-Criteria Decision Making (MCDM). MCDM methods are valuable when there is no clear and single "best" option and different decision criteria may conflict with each other. These methods help in structuring the decision problem, quantifying the criteria, and providing a systematic approach to rank or select alternatives based on their performance against the criteria.

In the present study two popular MCDM methods, AHP and TOSPIS are integrated for the ranking of various MLaaS providers. The Analytic Hierarchy Process (AHP) method is an extensively employed technique for MCDM. It is widely used in various fields such as business, agriculture, manufacturing industry, power generation, town planning healthcare, and more, to make informed decisions and to facilitate systematic comparisons between different options. It helps to structure complex problems, involves multiple stakeholders and provides consistent and rational solutions [2]. Another popular MCDM tool, TOPSIS was introduced by Hwang and Yoon in 1981. It enables the decision maker to select the best alternative depending upon a set of criteria, in a situation where many alternatives are available [3]. The TOPSIS method is particularly useful when there are conflicting criteria or trade-offs involved in the decision-making process. It takes into account both the positive and negative aspects of each alternative and provides a systematic way to compare and rank them. However, the method's effectiveness relies on the appropriate selection of criteria, accurate normalization, and meaningful weight assignment.

To rank various MLaaS service providers, an integrated approach of the AHP-TOPSIS method is adopted. The proposed study is organized into 6 sections. The introduction is followed by a literature survey. The methodology is explained in Section 3, where the working of the AHP and TOPSIS methods is explained. Section 4 discusses application of the hybrid approach of the AHP – TOPSIS method in evaluating the ranks of MLaaS providers

2. Literature Survey

The integration of MLaaS and MCDM can lead to powerful decision-making systems that leverage machine learning capabilities to enhance the decision-making process. Goraya compared various MCDM methods in cloud service selection and analyzed their complexity and robustness [4]. Also, a comparative study was made using PROMOTHEE, ELECTRE and TOPSIS for IaaS, or Infrastructure as a Service cloud service selection [5]. In a study, a comparative analysis of AHP, TOPSIS and PROMETHEE was performed in the evaluation of Cloud service providers [6]. In a similar study WSM, MOORA, TOPSIS, COPRAS and PROMETHEE methods were compared to rank various Cloud Service Providers [7].

Fuzzy AHP is applied to facilitate decision-makers in choosing appropriate cloud services [8]. Kumar in their study proposed the use of BWM and TOPSIS methods in determining the effectiveness of cloud services [9]. In a similar study by Mostafa, a Cloud service provider was selected using the Best-only method (BOM) [10]. TOPSIS and the Best Worst Method (BWM) were deployed [11]. Cloud service provider selection was made using fuzzy AHP and TOPSIS [12]. A novel TOPSIS (RE-TOPSIS) approach was proposed in the evaluation of various commercial platforms depending on the Quality of Service they deliver and the importance of each QoS to cloud

users [13]. Buyukozkan proposed a Fuzzy hybrid model of the Analytic hierarchy process, TOPSIS and VIKOR methods in the evaluation of cloud computing technology [14]. Nawaz, in a study, suggested Best Worst Method (BWM) and Markov chain-based integrated approach for ranking Cloud service providers [15].

In a study by Abdelaziz et al., Cloud Service Provider (CSP) ranking was proposed by an integrated approach of entropy technique, Single Valued Neutrosophic (SVN) and TOPSIS [16]. The adoption of Cloud Enterprise Resource Planning (CERP) was carried out using the Fuzzy AHP approach by Naveed et al. [17]. Robert, while analyzing MLaaS stressed upon the strategic selection of service platform [18]. Qin proposed the application of region-based reinforcement learning in performance improvement of the Internet of Things by optimizing Machine learning configurations [19]. Execution of machine learning as a service cloud on heterogeneous Graphics processing clusters leads to performance issues and thus needs changes in traditional scheduling techniques [20]. Kaymakci applied the AHP method in the selection of MLaaS for small and medium-scale manufacturing industries [21].

3. Methodology

To select the best MLaaS service provider, an integrated approach of the AHP and TOPSIS method is proposed in the present study, as shown in Fig.1. The proposed methodology has two phases. In the first phase, the AHP method will be used in the calculation of Criteria weight. The second phase includes a ranking of alternatives using the TOPSIS method. A brief introduction of AHP, and TOPSIS is presented in subsequent subsections.

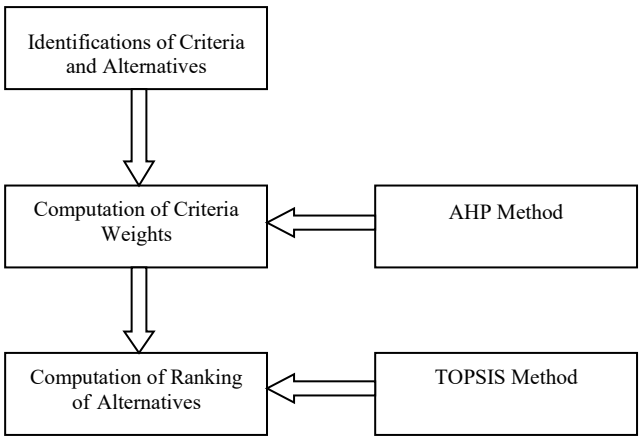


Fig. 1. The final Ranking of alternatives

3.1. Analytic Hierarchy Process (AHP)

The analytic Hierarchy Process (AHP) is a famous decision-making tool developed by Thomas L. Saaty[2]. The AHP method helps individuals or groups to make more rational decisions by breaking down the problem into a hierarchical structure, where the main problem is divided into a series of sub-problems and then further broken down into criteria. The AHP process involves the following steps:

Step 1: Goal Definition: Clearly define the main objective or goal of the problem under consideration.

Step 2: Criteria formulation: Identify the criteria that are relevant and important for evaluating the alternatives.

Step 3: Formulation of Hierarchical Structure: A hierarchical structure is designed where the overall goal of the problem forms the root of the tree. The next level is represented by criteria, sub-criteria and so on. The bottom-most level represents alternatives.

Step 4: Pairwise Comparison: In this step, decision-makers compare each element with another element to assign an importance value. This is done using a numerical scale to represent the relative importance (where 1 denotes equivalence and 9 denotes maximum importance).Satty's Scale, as Shown in Table 1 is used to make pairwise comparisons.

Table 1. Saaty's scale of comparison

Numerical Value	Definition
1	Two factors are of equal importance
3	Fist factor is some what more important
5	First factor is more important
7	First factor is very much important
9	First factor is absolutely more important
2,4,6,8	Intermediate values

Step 5: Normalize the criteria values to ensure that they are on the same scale. This is by converting the original values into a standardized scale, between 0 and 1. For criteria, that represent costs or negatives (minimization criteria), normalization will be reversed.

Step 6: Calculation of Priority Weights: Using the pairwise comparison data, the AHP method calculates the priority weights for each element. These weights reflect the relative weight that each component has in achieving the goal.

Step 7: Consistency Check: The consistency is checked to ensure the validity of the obtained priorities.
If the judgments are not consistent, decision-makers may need to reevaluate their pairwise comparisons.

3.2. TOPSIS Method

TOPSIS is a straightforward and effective method for decision-making when there are multiple criteria involved and it helps in determining the most suitable option based on the relative importance of each criterion [3]. It is based on the fact that best solution has the shortest path to ideal Solution, as shown in Fig.2.

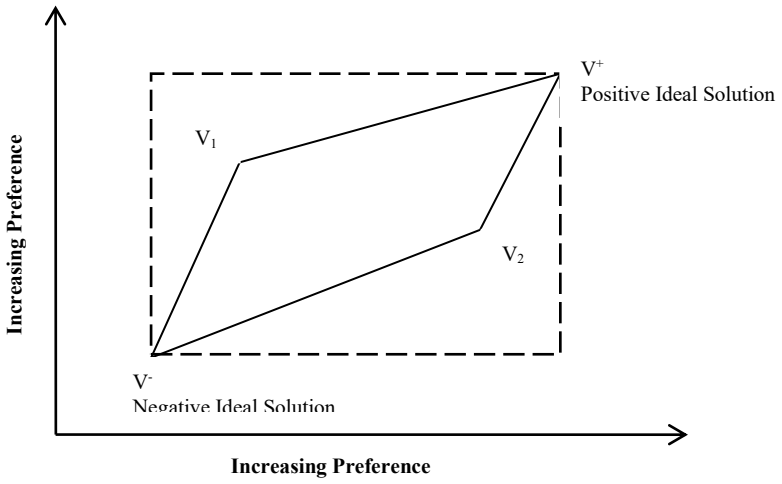


Fig. 2. The positive and negative ideal solutions in Topsis method

Euclidian distances are measured between best solution and ideal positive and negative solutions. However, it is essential to carefully define the decision matrix, criteria weights, and normalization process to obtain accurate and meaningful results. The steps of the TOPSIS method are as follows:

Step 1: Construction of Decision Matrix: A decision matrix can be represented as a 2D array or matrix, denoted as D. There are n alternatives (rows) and m criteria (columns) in the decision matrix. Where d_{ij} represents the value assigned to alternative i on criterion j.

Step 2: Construction of Normalized Decision Matrix: Normalize the values in the decision matrix to make them dimensionless and comparable across criteria. In the normalized decision matrix, each criterion is scaled to a range between 0 and 1, allowing for consistent comparisons across all criteria during the MCDM process.

$$\bar{X}_{ij} = \frac{X_{ij}}{\sqrt{\sum_{j=1}^{n} x_{ij}^2}} \quad (1)$$

Where X_{ij} is the value performance value in the ith row and j^{th} column of the Performance matrix.

Step 3: Calculation of weighted normalized decision matrix: Weights were applied to each criterion

Step 4: Determination of Ideal and Negative-Ideal Solutions: The maximum value of each criterion of the weighted Normalized decision matrix represents the best performance achievable on that criterion. Similarly, for cost (minimization) criteria, the minimum value corresponding to the criterion represents the best (lowest) cost achievable on that criterion.

Next, For each criterion, the negative-Ideal Solution is also computed. The minimum value for each criterion represents the worst performance possible on that criterion.
For cost (minimization) criteria, the maximum value in each criterion of the weighted normalized decision matrix represents the worst (highest) cost possible on that criterion.

Step 5: Calculation of Euclidean Distances: The Euclidean distance is a measure of the similarity or distance between two points in a multi-dimensional space and it represents the similarity or proximity of an alternative to the Ideal and Negative-Ideal solutions. Euclidian distance, S_j^+ was calculated using the following formula:

$$S_j^+ = \sqrt{\sum_{j=1}^{m} \left(V_{ij} - V_i^+\right)^2} \quad (2)$$

Similarly, Euclidian distance, S_j^- was calculated using the following formula:

$$S_j^- = \sqrt{\sum_{j=1}^{m} \left(V_{ij} - V_i^-\right)^2} \quad (3)$$

Step 6: Calculation of Relative Closeness: The relative closeness values indicate the relative preference of each alternative based on their proximity to the Ideal Solution and the negative-Ideal Solution. Performance Score Pi is calculated using the following formula

$$P_i = \frac{S_j^-}{S_j^+ + S_j^-} \quad (4)$$

4. Application of AHP - TOPSIS method in ranking of MLaaS providers

In this section step by step-by-step evaluation process of ranking of MLaaS providers is performed. Though many MLaaS providers are operating in the market[22][23][24], however, the present study is considering the following four most popular Machine Learning as a Service provider:

- •Amazon SageMaker (AWS)
- •Google Cloud AI Platform
- •Microsoft Azure Machine Learning

•IBM Watson Machine Learning

The detailed steps of the evaluation process are as follows:

Step 1: Selection of criteria
Seventeen criteria are selected: Anomaly Detection, Built-In Algorithms, Clustering, Cost Effectiveness, Customization Options, Deployment, Filtering Inappropriate Content, Flexibility, Interoperability, Metadata extraction, Pipeline Support, Ranking, Recommendation, Relations analysis, Supported framework on cloud, Usability, User friendliness.

Step2: Pairwise Comparison
Each criterion is compared with every other criterion. The relative importance with respect to the criteria is assigned using Satty's Scale (Table 1), a numerical scale to represent the relative importance from 1 to 9. The resultant matrix is shown in Table 2.

Table 2. Pairwise comparison matrix

Criteria	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17
C1	1	0.33	2.00	0.33	0.33	0.33	1.00	0.33	0.33	1.00	0.50	2.00	2.00	2.00	0.33	0.33	0.33
C2	3.00	1	2.00	0.50	0.50	0.50	3.00	0.50	0.50	3.00	2.00	2.00	2.00	2.00	1.00	0.50	0.50
C3	0.50	0.50	1	0.33	0.33	0.33	0.50	0.33	0.33	0.50	0.50	1.00	1.00	1.00	0.33	0.33	0.33
C4	3.00	2.00	3.00	1	3.00	3.00	3.00	2.00	3.00	2.00	2.00	3.00	3.00	3.00	2.00	3.00	2.00
C5	3.00	2.00	3.00	0.33	1	1.00	3.00	0.50	1.00	4.00	3.00	3.00	3.00	3.00	2.00	1.00	0.50
C6	3.00	2.00	3.00	0.33	1.00	1	4.00	0.50	1.00	4.00	3.00	3.00	3.00	3.00	2.00	1.00	0.50
C7	1.00	0.33	2.00	0.33	0.33	0.25	1	0.33	0.33	1.00	0.50	2.00	2.00	2.00	0.33	0.33	0.33
C8	3.00	2.00	3.00	0.50	2.00	2.00	3.00	1	2.00	4.00	3.00	4.00	4.00	4.00	3.00	2.00	0.50
C9	3.00	2.00	3.00	0.33	1.00	1.00	3.00	0.50	1	4.00	3.00	3.00	3.00	3.00	2.00	1.00	0.50
C10	1.00	0.33	2.00	0.50	0.25	0.25	1.00	0.25	0.25	1	0.50	2.00	2.00	2.00	0.33	0.25	0.33
C11	2.00	0.50	2.00	0.50	0.33	0.33	2.00	0.33	0.33	2.00	1	2.00	2.00	2.00	0.50	0.33	0.50
C12	0.50	0.50	1.00	0.33	0.33	0.33	0.50	0.25	0.33	0.50	0.50	1	1.00	1.00	0.33	0.33	0.33
C13	0.50	0.50	1.00	0.33	0.33	0.33	0.50	0.25	0.33	0.50	0.50	1.00	1	1.00	0.50	0.33	0.25
C14	0.50	0.50	1.00	0.33	0.33	0.33	0.50	0.25	0.33	0.50	0.50	1.00	1.00	1	0.50	0.33	0.33
C15	3.00	1.00	3.00	0.50	0.50	0.50	3.00	0.33	0.50	3.00	2.00	3.00	2.00	2.00	1	0.50	0.50
C16	3.00	2.00	3.00	0.33	1.00	1.00	3.00	0.50	1.00	4.00	3.00	3.00	3.00	3.00	2.00	1	0.50
C17	3.00	2.00	3.00	0.50	2.00	2.00	3.00	2.00	2.00	3.00	2.00	3.00	4.00	3.00	2.00	2.00	1

Step3: Calculation of Priority Weights
Using the pairwise comparison data, the AHP method calculates the priority weights for each element of the hierarchy. An online software tool by Gospel [25] was used to make comparisons and weight computation. A total 136 number of comparisons were made to calculate weights for criteria. Weights for criteria are shown in Table 3.

Table 3. Criteria Weights

S.No	Criteria	Weight
C1	Anomaly Detection	4.1
C2	Built-In Algorithms	5.5
C3	Clustering	3.1
C4	Cost Effectiveness	11
C5	Customization Options	7.5
C6	Deployment	8.0
C7	Filtering Inappropriate Content	3.8
C8	Flexibility	9.5
C9	Interoperability	7.6
C10	Metadata extraction	4.2
C11	Pipeline Support	4.1
C12	Ranking	3.1
C13	Recommendation	3.1
C14	Relations analysis	3.1
C15	Supported framework on cloud	5.7
C16	Usability	7.5
C17	User friendliness	9.1

Step 4: Consistency Check: The consistency ratio was computed to ensure the validity of the obtained priorities. Here, Consistency Ratio CR = 3.6%. Thus weight computation is consistent.

Step 5: Formation of Decision matrix:
A decision matrix is a table that represents a set of alternatives evaluated on multiple criteria. It is used to systematically compare and rank the alternatives based on their performance with respect to the criteria. The decision matrix is shown in Table 4.

Table 4. Decision Matrix

Weight.	4.1	5.5	3.1	11	7.5	8	3.8	9.5	7.6	4.2	4.1	3.1	3.1	3.1	5.7	7.5	9.1
Criteria	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	12	C13	C14	C15	C16	C17
ALT 1	5	3	5	2	3	5	0	5	5	0	4	5	5	0	5	4	4
ALT2	0	3	5	5	5	3	5	2	5	0	4	0	5	0	4	4	5
ALT3	0	0	0	3	3	5	0	4	4	5	5	0	0	5	3	5	5
ALT4	5	0	5	3	5	3	5	4	4	0	5	5	5	5	4	5	3

Step 6: Normalized and Weighted normalized decision matrix:
Vector normalization was done using eq (1) to get values in each cell of the normalized decision matrix. Next, each criterion's weight is multiplied by each cell's normalized performance value in a normalized decision matrix.

Step 7: Calculation of ideal best V_j^+ and ideal worst value V_j^- for all criteria.
The largest value in that row will be the ideal best, while the least value will be the ideal worst. Table 5 shows a weighted normalized decision matrix with V_j^+ and Vj^- calculations.

Table 5. Weighted normal matrix

Alternatives➔ Criteria⬇	Amazon SageMaker	Google Cloud AI Platform	IBM Watson Machine Learning	Microsoft Azure Machine Learning	V^+	V^-
Anomaly Detection	2.899138	0.000000	0.00000	2.899138	2.899138	0.00000
Built-In Algorithms	3.889087	3.889087	0.00000	0.00000	3.889087	0.00000
Clustering	1.789786	1.789786	0.00000	1.789786	1.789786	0.00000
Cost Effectiveness	3.20903	8.022575	4.813545	4.813545	8.022575	3.20903
Customization Options	2.728526	4.547543	2.728526	4.547543	4.547543	2.728526
Deployment	4.850713	2.910428	4.850713	2.910428	4.850713	2.910428
Filtering Inappropriate Content	0.000000	2.687006	0.000000	2.687006	2.687006	0.00000
Flexibility	6.081752	2.432701	4.865401	4.865401	6.081752	2.432701
Interoperability	4.196398	4.196398	3.357118	3.357118	4.196398	3.357118
Metadata extraction	0.00000	0.00000	4.200000	0.00000	4.200000	0.00000
Pipeline Support	1.811077	1.811077	2.263846	2.263846	2.263846	1.811077
Ranking	2.192031	0.00000	0.00000	2.192031	2.192031	0.00000
Recommendation	1.789786	1.789786	0.00000	1.789786	1.789786	0.00000
Relations analysis	0.000000	0.00000	2.192031	2.192031	2.192031	0.00000
Supported framework on cloud	3.508107	2.806486	2.104864	2.806486	3.508107	2.104864
Usability	3.312946	3.312946	4.141182	4.141182	4.141182	3.312946
User friendliness	4.203110	5.253887	5.253887	3.152332	5.253887	3.152332

Step 9: Calculation of Euclidian distance : Euclidian distance, S_j^+ was calculated using eq(2). Similarly, Euclidian distance, S_j^- was calculated using eq (3).

Step 10: Calculation of Performance Score P_i: Performance Score P_i is calculated using eq (4) Finally, based on the performance score, alternatives were ranked. The option with the highest value of P_i will be ranked first, indicating the most preferred alternative. The final ranks of alternatives are shown in Table 6.

Table 6. Rank Computations

Alternatives	Si+	Si-	Pi	Rank
Amazon SageMaker	7.624903	7.456511	0.494417	2
Google Cloud AI Platform	7.57112	8.002951	0.513864	1
IBM Watson Machine Learning	7.727959	6.325323	0.450096	4
Microsoft Azure Machine Learning	7.354052	6.69524	0.476554	3

5. Results and Analysis

Machine learning as a service over the cloud is offered by many cloud service providers. However, four leading service providers were selected for the study. MLaaS from Amazon, Google, IBM and Microsoft constitute the alternatives. These alternatives were evaluated based on seventeen criteria: Anomaly Detection, Built-In

Algorithms, Clustering, Cost Effectiveness, Customization Options, Deployment, Filtering Inappropriate Content, Flexibility, Interoperability, Metadata extraction, Pipeline Support, Ranking, Recommendation, Relations analysis, Supported framework on cloud, Usability, User friendliness.

Criteria were compared using the AHP method and the weight corresponding to each criterion was calculated. TOPSIS method was then applied for the calculation of ranks. The ranking of alternatives is shown in Fig 3.

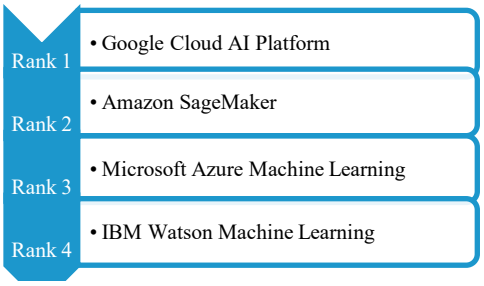


Fig. 3. The final Ranking of alternatives

Google Cloud AI Platform is ranked one as its Performance Score Pi, 0.513864 is the highest. It is followed by Amazon SageMaker. IBM Watson Machine Learning is last in the ranking as its performance score is least.

6. Conclusion and Future Research Directions

Cloud computing has transformed the IT landscape by offering scalable, cost-effective, and accessible solutions for individuals, businesses, and organizations. The present study proposed a multi-criteria-based approach to compare and rank various Machine Learning as Service over cloud service providers. The integrated approach of The AHP and TOPSIS method offers a transparent approach for evaluating different MLaaS (Machine Learning as a Service) providers. However, determining criteria weights using The AHP method can be subjective and may lead to potentially biased results.

The MLaaS market is dynamic, with new service providers, technologies, and features emerging regularly. Conducting an MCDM evaluation might require frequent updates to ensure the analysis remains relevant and accurate. Some MLaaS providers might have unique features or offerings that are difficult to incorporate into a standardized evaluation framework. These specific factors might be critical for a particular organization, but not adequately captured in the general MCDM approach. Some criteria were challenging to quantify objectively. Thus, The concepts from Fuzzy set theory can be incorporated to handle such criteria. Also, the involvement of various stakeholders will help to achieve more objective criteria weighting and decision-making.

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